

Unified Computational Framework for High-Fidelity CFD, Combustion Kinetics, and Optimal Orbital Trajectories

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Abstract

This paper presents a unified computational environment for the simulation and optimization of aerospace systems. We integrate high-fidelity quasi-1D CFD nozzle modeling with Arrhenius combustion kinetics, Pontryagin-based optimal throttle control, and 2-D gravity-turn trajectory analysis. Each module is grounded in rigorous mathematical formulations, ranging from non-linear PDEs to linearized finite-math state-space representations. Our results demonstrate the scalability of these models for real-time mission control and structural thermal analysis of regenerative cooling systems.

1. Advanced CFD and Thermal Modeling

The gas dynamics within the rocket nozzle are resolved using the quasi-1D Navier-Stokes equations with area variation (A). Convective heat flux (q_{wall}) is modeled using the Bartz equation:

$$\begin{aligned} h_g &= [0.026 / D^{0.2}] * [\mu^{0.2} * Cp / Pr^{0.6}] * [Pc / C^*]^{0.8} * [Dt / r]^{0.1} \\ q_{wall} &= h_g * (T_{gas} - T_{wall}) \\ dT_{wall}/dt &= q_{wall} / (\rho_w * Cp_w * \delta_w) \end{aligned}$$

2. Combustion PDE and Arrhenius Kinetics

Flame front evolution is resolved through coupled species and energy transport equations:

$$\begin{aligned} dT/dt &= \alpha * d^2T/dx^2 + Q * w / (\rho * Cp) \\ dY_f/dt &= D * d^2Y_f/dx^2 - w / \rho \\ w &= A * (\rho * Y_f) * (\rho * Y_o) * \exp(-E_a/RT) \end{aligned}$$

3. Optimal Throttle via Pontryagin's Principle

Throttle optimization minimizes the fuel functional $J = \int (|u(t)|) dt$ subject to dynamics:

$$\begin{aligned} dv/dt &= [T_{max} * u / m] - g(h) - [D(v, h) / m] \\ dm/dt &= - [T_{max} * u / (Isp * g_0)] \end{aligned}$$

4. 2-D Gravity-Turn Trajectory Analysis

Vehicle motion is integrated in a non-rotating coordinate system with varying gravity:

$$\begin{aligned} dx/dt &= v * \cos(\gamma) \\ dz/dt &= v * \sin(\gamma) \\ dv/dt &= [T - D]/m - g * \sin(\gamma) \end{aligned}$$

$$d(\gamma)/dt = [-g * \cos(\gamma)] / v$$

5. Mass Budget and Tsiolkovsky staging

The multi-stage rocket capability is analyzed using the cumulative delta-v budget:

$$\Delta v_{\text{total}} = \sum [I_{sp_i} * g_0 * \ln(m_{\text{initial}_i} / m_{\text{final}_i})]$$

6. Finite Math and State-Space Stability

Linearized pitch-plane stability is assessed using the state transition matrix $\Phi(t)$:

$$dx/dt = Ax + Bu$$

$$\Phi(t) = \exp(At) = \text{Inv_Laplace} \{ (sI - A)^{-1} \}$$

$$\text{Stability} \Leftrightarrow \max(\text{Re}(\text{eig}(A))) < 0$$

7. Conclusion

The integrated computational suite provides a comprehensive toolset for aerospace vehicle design. By bridging the gap between microscopic combustion PDEs and macroscopic orbital trajectories, we enable holistic optimization of space-faring systems.